

Homework - Find the Biggest Value

This is a programming assignment that will be graded and scored as homework since it is so simple.

Write a program for the LC3. This program should be named ***hwBigSwap.hex***. The file should be a proper hex file for the LC3 processor.

A proper hex file is a text file that contains only hex characters (0-9 and A-F).

Each line should be a four-digit hex number. Do not include a 0x or x at the beginning of this number.

The first line should be the starting address in memory, use 3000 to be safe.

The last line should be a halt (F025)

DO NOT add any extra spaces at the end of lines.

BE CAREFUL not to use the O (oh) in place of the 0 (zero) or the l (ell) in place of the 1 (one).

The lines between the first and last line should do the following:

Examine the values in R0 and R1.

Make sure the value in R0 is greater than the value in R1.

If R0 is less than R1, swap the values in R0 and R1.

Do not initialize R0 and R1, simply assume the values have already been placed in these registers.

Write your program in assembly. Convert it manually to hex. Put the hex in hwBigSwap.hex and upload.

Submit your hex file to Web-CAT. Make sure you or your text editor did not add **.txt** to the end of the filename.

Testing your program

1. Open your program in the simulator but don't run it.
2. Click on R0 and put the value 9 in it.
3. Click on R1 and put the value 5 in it.
4. Run your program. It should not swap anything.
5. R0 should contain 9 and R1 should contain 5.
6. Reset your program.
7. Click on R0 and put the value 5 in it.
8. Click on R1 and put the value 9 in it.
9. Run your program. It should swap the values in R0 and R1.
10. R0 should now contain 9 and R1 should now contain 5.
11. Repeat with different values.
12. Try zero in R0. Try zero in R1. Try zero in R0 and R1. Try negative numbers. After running your program, the largest value should always be in R0.